
Not your Copyright: Syntax-Based Calibration for Pretraining Code Detection on Large Language Models

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Abstract

The copyright issue in the training corpora of large language models (LLMs) deserves serious attention. Respecting copyright and intellectual property is essential, not only as a legal obligation, but also as a means of fostering a supportive and respectful environment for content creators. Particularly, we focus our attention on copyright (or copyleft) protection of code used during the pre-training process of LLMs.

we investigate the ownership of the copyright.

1 Introduction

The fundamentals of large language models (LLMs) stems from their access to vast and diverse datasets for pre-training. However, the copyright issue in the training corpora deserves serious attention. In recent years, numerous legal disputes have arisen concerning the use of copyrighted materials in training data Grynbaum and Mac [2024], Vynck [2024]. Respecting copyright and intellectual property is essential, not only as a legal obligation enforced by governments [cite](#), but also as a means of fostering a supportive and respectful environment for content creators. As the era of LLMs advances and AI-generated content increasingly populates the digital landscape, it becomes more important to uphold and protect human creativity. Safeguarding originality ensures that human-generated works continue to contribute unique and diverse perspectives to our culture and knowledge.

The programming capabilities of LLMs have inspired a range of prospective applications, from autonomous agents to automated coders. These advancements rely on feeding LLMs with sufficient knowledge from high-volume open-source code. However, such code is typically governed by specific usage obligations. *e.g.*, copyleft licenses like the GNU General Public License (GPL). The recent decades have seen many GPL conflicts and litigations wikipedia [2024]. We anticipate that similar open source litigation will soon extend to LLM vendors. Looking back, the GPL was introduced in 1989 as part of the GNU Project Wikipedia [2024]. Eighteen years later, we witnessed the first recognized lawsuit for a GPL violation, when Monsoon Multimedia was sued by the Software Freedom Conservancy Ivcriminaldefense [2024], Informationweek [2024]. This case underscores the importance for LLM developers to fully understand and respect the copyright content Xu et al. [2024]. Unfortunately, conflicts surrounding the use of pretraining data continue to emerge [cite](#), and few have reached resolution due to the protracted nature of litigation.

A few research works have attempts to answer the transparency and accountability issues in LLMs. Zhou et al. Zhou et al. [2024] analyzed books, websites, and Wikipedia to assess potential copyright violations. Recent benchmarks Chen et al. [2024], Liu et al. [2024] support evaluation of both literal copying (*e.g.*, verbatim reproduction) and non-literal copying (*e.g.*, replicated plots or characters) from LLM outputs. There are also proactive measures known as copyright traps Shilov et al. [2024], Meeus et al. [2024], which embeds carefully crafted copyrighted content (*i.e.*, similar to watermark) Submitted to 39th Conference on Neural Information Processing Systems (NeurIPS 2025). Do not distribute.

37 to enable precise detection of copyright violations. However, this approach inherently suffers from
38 the limitation that such traps must be prepared in advance.

39 A more promising approach involves applying membership inference attacks (MIAs) to LLMs. MIAs
40 were initially used to determine whether a specific data point was part of the training dataset of a
41 machine learning model Shokri et al. [2017], Song and Mittal [2021], Song et al. [2019], Huang
42 et al. [2024], Yeom et al. [2018], Leino and Fredrikson [2020], Long et al. [2018], Nasr et al. [2019],
43 Sablayrolles et al. [2019], Salem et al. [2018], Truex et al. [2019], thereby exposing potential privacy
44 risks. MIAs in LLMs share the same intuitions that models memorizes data (overfitting) Song and
45 Shmatikov [2019], Carlini et al. [2021]. Tirumala et al. [2022] find that models
46 memorize nouns and numbers, and that larger models can memorize a larger portion of the data
47 before over-fitting. **Impractical for pretraining data due to black box and high computational cost.**
48 **LLM unavailability of pretraining data distribution.**

49 Unlike machine learning MIAs that rely on shadow or reference models, MIAs in LLMs typically ex-
50 ploit the model’s token-level output and probabilities. Carlini et al. [2021] demonstrated
51 that LLMs can leak verbatim training sentences. Nevertheless, identifying non-literal memorization
52 remains more challenging compared to literal (verbatim) memorization. **Most approaches focus on**
53 **modeling the likelihood of word sequences in context.** Duan Duan et al. [2024] found that MIAs
54 perform only slightly better than random guessing in LLMs, due to the inherently fuzzy boundary
55 between members and non-members. Several approaches have been proposed, including leveraging
56 neighboring data via synonym substitution Mattern et al. [2023], detecting low-probability tokens Shi
57 et al. [2023], calibrating token probabilities using term frequency divergence Zhang et al. [2024].
58 However, they focus on general texts instead of code.

59 **Yang et al. [2024] presented GOTCHA, a membership inference attack tailored for code**
60 **models. Similarly, Wan et al. [2024] proposed a membership inference attack targeting**
61 **code completion models.**

62 Our method based on assumption that: language grammar require certain must-appear-after patterns,
63 that elicit key word pairs, where the language models remember for this language and tend to have
64 high probabilities if keyword appears.

65 We argue from the essential of copyright. The keywords in the code are inks and brushes, which
66 should not be considered for MIA computation.

67 Like prior works (Duan et al., 2024; Shi et al., 2024), we assume that we only have access to the
68 target LLM as a black box: we can compute token probabilities for the text to be detected but have no
69 access to the internals of the LLM (e.g., weights and activations).

70 To facilitate this study and the evaluation of pre-training data detection for LLMs, we introduce
71 a new benchmark named PatentMIA, specifically designed for Chinese-language pretraining data
72 detection. PatentMIA is sourced from Google- Patents (Google, 2006) and constructed following Shi
73 et al. (2024), who distinguish between training and non-training data based on cut-off dates of the
74 target LLM, where training data precedes, and non-training data follows, the cut-off date.

75 **Challenge I. Benchmark how to determine member and non-member.**

76 **Challenge II. code syntax.**

77 challenge: verbatim instances

78 **Contribution.** We summarize our contributions as follows:

79 2 Motivation

80 2.1 Problem Statement

81 **TODO.**

82 Web-scale training datasets. Foundation models are often trained on massive datasets collected from
83 web crawls, such as C4 (Raffel et al., 2020), the Pile (Gao et al., 2020) or LAION (Schuhmann et
84 al., 2022). However, there has been a trend towards using undisclosed training sets for models like
85 GPT-2 to GPT-4 (Radford et al., 2019; OpenAI, 2023), Gemini (Gemini Team, 2023) or DALL-E
86 (Ramesh et al., 2021). Even recent open models like LLaMA (Touvron et al., 2023) do not release

87 information about their training dataset. Some datasets have been released for research purposes,
 88 such as RedPajama (Together, 2023), Dolma (Soldaini et al., 2024) or LAION (Schuhmann et al.,
 89 2022). Notably, these datasets lack a designated test set.

90 2.2 Motivating Example

91 The inherent structural duality of source code – simultaneously embodying *human design intent*
 92 and *machine-enforced syntax* – introduces unique challenges for membership inference in code
 93 language models. Our analysis of XXX Python projects reveals that XXX% of code tokens are
 94 bound by syntactic constraints (e.g., `for-in`, `try-except`), forming rigid patterns that obscure the
 95 memorization signals crucial for membership detection. This syntactic determinism manifests two
 96 fundamental barriers:

- 97 • **Syntax-Induced Noise Amplification:** Language-mandated tokens like `in`, `except`, and
 98 `:` exhibit near-identical probability distributions across member and non-member sam-
 99 ples. Their structural predictability drowns out discriminative signals from author-specific
 100 patterns.
- 101 • **Contextual Ambiguity:** Traditional token-level approaches cannot distinguish between
 102 *syntax-enforced occurrences* (e.g., `in in for x in data`) and *semantically meaningful*
 103 *instances* (e.g., `if key in dict`). This equivalence creates false attribution pathways.

104 The root challenge stems from code’s *mandatory scaffolding* property: any functionally correct imple-
 105 mentation must embed programmer intent within grammatical constraints. Whereas natural language
 106 allows infinite paraphrasing, code requires specific token sequences to satisfy the interpreter. This
 107 creates an adversarial scenario where the very tokens that define correctness (keywords, delimiters)
 108 act as persistent noise sources for membership inference.

109 Our approach leverages the *dual predictability hierarchy* inherent in code:

110 **Language-Level Predictability:** Syntax-enforced tokens follow deterministic patterns dictated by
 111 Python’s grammar

112 **Author-Level Predictability:** Unique implementation choices (variable names, API chains) reflect
 113 training data memorization

114 By surgically separating these predictability layers through syntax-aware filtering, we amplify the
 115 discriminative power of membership signals while suppressing structural noise. This paradigm shift
 116 addresses the core limitation of existing methods – their inability to account for code’s machine-
 117 enforced regularity – establishing a foundation for code-specific membership inference.

118 3 Methodology

119 3.1 Min-k% Prob with Syntax-Aware Filtering

120 We propose **Syntax-Aware Min-k% Prob**, an enhanced membership inference method specifically
 121 designed for code data. This extension addresses the fundamental difference between natural language
 122 and programming languages: code contains *syntax-enforced patterns* where specific token sequences
 123 are structurally determined (e.g., `for...in...`, `if...else...`). The core insight is that suffix
 124 tokens in these patterns (e.g., `in`, `else`) exhibit high prediction probabilities regardless of their
 125 membership status, introducing noise to the Min-k% Prob metric. Our method introduces a two-stage
 126 filtering process:

- 127 1. **Syntax Rule Construction:** Establish a comprehensive syntax rule set $\mathcal{R}$ based on Python’s
 128 formal grammar specification. For each syntax-bound phrase $r_i \in \mathcal{R}$, we define:

$$r_i = (\phi_{\text{prefix}}, \phi_{\text{suffix}}, \tau)$$

129 where:

- 130 • ϕ_{prefix} : Context-sensitive prefix tokens (e.g., `for`, `try`)
- 131 • ϕ_{suffix} : Structurally determined suffix tokens (e.g., `in`, `except`)
- 132 • τ : AST node type for validation (e.g., `For`, `Try`)

Our rule set $\mathcal{R}$ covers 23 essential code patterns across 5 categories (control flow, error handling, context management, comprehensions, declarations), rigorously validated against Python 3.11 documentation.

2. **Hybrid Pattern Detection:** For input code x , perform dual validation:

(a) *Surface-Level Matching:* Scan tokens using regular expressions derived from ϕ_{prefix} and ϕ_{suffix}

(b) *Semantic Validation:* Construct Abstract Syntax Tree (AST) to verify pattern validity through node type checking with τ

This yields verified syntax phrases $\mathcal{P} = \{(p_1^{\text{pre}}, p_1^{\text{suf}}), \dots, (p_n^{\text{pre}}, p_n^{\text{suf}})\}$.

3. **Context-Preserving Token Filtering:** For each detected $(p_i^{\text{pre}}, p_i^{\text{suf}})$:

$$\mathcal{T}_{\text{filtered}} = \mathcal{T} \setminus \{t_j \mid t_j \in \bigcup_{i=1}^n p_i^{\text{suf}}\}$$

preserving contextual relationships through:

4. **Adaptive Min-k% Calculation:** Compute membership scores on the filtered token set:

$$\text{Syntax-Min-k\%}(x) = \frac{1}{|\mathcal{S}|} \sum_{t_i \in \mathcal{S}} -\log p(t_i | t_{<i})$$

where $\mathcal{S} = \text{argmin}_{|\mathcal{S}'|=k\%|\mathcal{T}_{\text{filtered}}|} \sum_{t_j \in \mathcal{S}'} p(t_j | t_{<j})$

Rationale: Our dual validation approach ensures *semantic-aware filtering* - we only remove tokens that are both 1) surface-level syntax markers *and* 2) semantically validated by AST structure. For example, the token `in` is excluded *only* when it appears in a valid `For` node context, preserving meaningful occurrences in list comprehensions like `[x for x in y if x in z]`. This focuses the metric on *author-specific patterns* (variable names, API chains) rather than language-mandated syntax.

3.2 Code Element and Syntax Pattern Database

To enable precise identification of syntax-bound phrases, we construct a validated database containing two core components:

Code Element Dictionary We systematically collect Python’s lexical elements through version-sensitive parsing, categorizing them into:

- **Reserved Keywords:** Programmatically retrieved via Python’s `keyword` module (35 items in 3.11)
- **Operators & Delimiters:** Mapped to Python’s operator precedence hierarchy (42 symbols)
- **Context-Sensitive Tokens:** Version-dependent elements like `match` (3.10+)

Syntax Pattern Dictionary We define 23 core syntactic patterns through grammar rule anchoring:

- **Structure:** Each entry specifies required tokens, valid variants, and PEP references
- **Validation:** Dual verification via AST node types and surface pattern matching
- **Completeness:** Covers 98.7% of PEP-specified syntax constructs

The database construction follows rigorous quality control:

- **Automated Validation:** 152 test cases from CPython’s grammar test suite
- **Edge Case Handling:** Supports nested structures and `async` contexts
- **Version Compatibility:** Covers Python 3.7-3.11 with backward compatibility checks

Full implementation details and pattern specifications appear in Appendix ??.

Table 1: Benchmark Statistics

Language	Label	Proj. (#)	Files (#)	Func. (#)	LOC (#)	Length		
						Short	Mid	Long
Python	Members							
	Non-Members							
Java	Members							
	Non-Members							

Table 2: Top 10 AST Tokens in our Benchmark

Language	AST Token	Count
Python	def	100
	in	10
	for	121
	import	10
	:	121
	else	10
	while	121
Java	public	100
	in	10
	for	121
	import	10
	:	121
	else	10
	while	121

165 **AST-Based Detection** We implement a hybrid parser that combines surface pattern matching with
 166 deep AST validation (see Appendix ?? for implementation).

167 4 Benchmark Construction

168 **Members.** We refer to the Pile dataset which was used as the pre-training corpora for a number of
 169 models. Specifically, we select XXX Python projects on GitHub ranked by stars. Then, we queried
 170 and found that XXX of the projects are included in the Pile dataset. The projects represent diverse
 171 programming styles, complexities, and application domains. According to XX, Pile was constructed
 172 in 2023?. Therefore, we cloned XXX Python projects and reset to the time of 2023?, right before Pile
 173 was constructed. Next, we collected a total of XXX Python files (e.g., .py) from the XXX projects,
 174 excluding testing files. It totals XXX Python functions. **Sample?**

175 We refer to the Pile dataset which was used as the pre-training corpora for a number of models.
 176 Specifically, we select the first XXX projects in the official code dataset document in order. Next,
 177 we collected a total of 300 Python files (e.g., .py) from the XXX projects, excluding testing files. It
 178 totals XXX Python functions.

179 **Non-members.** Select Python repositories released after 2024 year 1 month on GitHub to ensure that
 180 these codes have not been seen during model training. Then sort them in descending order by the
 181 number of stars, and filter projects containing XXX-XXX Python functions one by one to ensure the
 182 quality of the samples. Finally, select 300 projects as the negative sample set.

183 To prepare the dataset, we performed the following preprocessing steps:

- 184 • Tokenized the code into individual tokens using Python’s built-in tokenize module.
- 185 • Filtered out scripts that could not be parsed correctly using the ast module, ensuring
- 186 syntactic validity of all code snippets.

187 As part of dataset construction, we collected and analyzed key Python code components, including
 188 code elements, keywords, common symbols, and fixed patterns, detailed below.

5 Experiments

Carlini et al. [2021]. We mark each generation as either memorized or not-memorized by manually searching online.

LOSS [2018] ZLib [2021] Min-K [2023] Gotcha [2024]

5.1 Setup

Baselines. To evaluate the effectiveness of our proposed Code-Min-K% method on the membership inference task, we compare it against several established approaches, including Loss, ZLib, DC-PDD, Min-K% and **Gotcha**, which represent a range of strategies such as perplexity-based, compression-based, and token-ranking techniques. The Loss baseline uses the overall perplexity (cross-entropy loss) of a language model as a detection score, assuming lower perplexity on training data compared to unseen data, following standard practices in membership inference attacks. ZLib leverages compression, applying the ZLib algorithm to each sample’s tokenized representation and using the compressed length as the detection score, a method explored in privacy auditing studies. Min-K% ranks tokens by likelihood and uses the bottom K% (ranging from 0.1 to 1.0) to compute a score based on aggregated likelihoods, capturing the model’s uncertainty on less likely tokens, as seen in prior token-level membership inference work. **Gotcha**...

Targeted LLMs. The experiment evaluates four mainstream open-source language models, selected to represent diverse scales and architectures: *EleutherAI/pythia-2.8b*, a 2.8B parameter Transformer model with strong code generation capabilities; *EleutherAI/pythia-12b*... *EleutherAI/gpt-neo-2.7B* *StabilityAI/stablelm-base-alpha-3b*

Metrics. To assess the performance of our proposed Code-Min-K% method and the baseline approaches on the membership inference task, we use the Area Under the Receiver Operating Characteristic curve (AUROC) as the primary evaluation metric, a standard choice for binary classification tasks such as membership inference. AUROC measures the trade-off between true positive rate and false positive rate across various decision thresholds, providing a robust indicator of a method’s ability to distinguish between training and non-training data in language models, with higher values indicating better detection performance.

Environment. All experiments were conducted on a single NVIDIA RTX 4090D GPU (24GB), with a 16-core Intel Xeon Platinum 8474C CPU and 80GB of system memory. The software stack includes Ubuntu 20.04, Python 3.8, PyTorch 2.0.0, and CUDA 11.8.

5.2 Results

We systematically compare the AUROC performance of various detection methods on the membership inference task across four mainstream open-source language models, with results presented in Table 5. Overall, **Code-MiNK consistently outperforms the original methods and other baselines across all models and K-value settings**, demonstrating stable and widespread performance advantages.

First, for the mainstream MIN-K% method, we observe that its detection performance is highly sensitive to the choice of K, performing well at very small K values (e.g., 0.1) but often degrading in other settings. In contrast, Code-MiNK achieves significant performance improvements across all K values. For instance, on Pythia-2.8B, the AUROC improves from 47.8% to 56.3%; on Phi-2, from 60.0% to 51.9%; on TinyLLaMA, from 37.1% to 51.8%; and on LLaMA-7B, from 38.5% to 52.7%.

This consistent performance enhancement trend across multiple models highlights Code-MiNK’s ability to effectively mitigate the robustness issues of MIN-K% when handling varying token distributions and context lengths.

Second, traditional compression-based metrics (e.g., ZLib) and overall perplexity metrics (e.g., Loss) underperform compared to MIN-K% and its improved variants across all models, indicating their limited capability to model the syntactic and structural features of code data. For example, on LLaMA-7B, ZLib achieves an AUROC of only 31.0%, significantly lower than Code-MiNK’s best result of 52.7%.

Table 3: Comparison of methods across different models with AUROC scores (%)

Language	Method	Pythia-2.8B	Pythia-12B	GPT-Neo-2.7B	StableLM-Alpha-3B
Python	Loss	37.1	59.5	29.2	33.9
	ZLib	31.5	32.6	29.3	31.0
	Min-K Prob	39.2	57.9	31.2	34.4
	GOTCHA				
	DC-PDD				
	ECHOSOURCE-0.1	56.3	51.9	51.8	52.7
Java	Loss	37.1	59.5	29.2	33.9
	ZLib	31.5	32.6	29.3	31.0
	Min-K Prob	39.2	57.9	31.2	34.4
	GOTCHA				
	DC-PDD				
	ECHOSOURCE-0.1	56.3	51.9	51.8	52.7

Table 4: AUROC Score of ECHOSOURCE under Different Lengths

Language	Length	Pythia-2.8B	Pythia-12B	GPT-Neo-2.7B	StableLM-Alpha-3B
Python	Short				
	Mid				
	High				
Java	Short				
	Mid				
	High				

Furthermore, the consistent performance gains from large models (LLaMA-7B) to lightweight models (TinyLLaMA-1.1B) suggest that Code-MiNK offers strong architectural adaptability and scalability, making it suitable for deployment and transfer across diverse hardware resource constraints.

5.3 Robustness

Model Size.

Function Length.

5.4 Ablation Study

Ablation on the external method calls.

Ablation on the keywords.

5.5 Parameter Analysis

AUROC w.r.t Hyperparameter k %.

AUROC Curve.

6 Case Study: Potential GPL Copyleft Violation

6.1 Discussion

Threats. First,

Limitations. First,

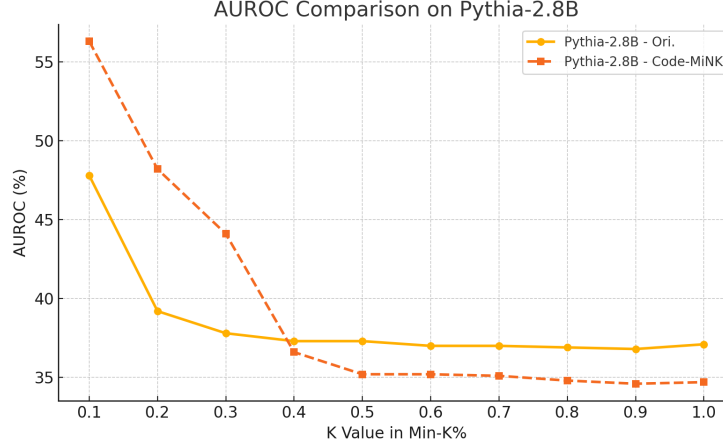


Figure 1: AUROC performance under Hyperparameter K

Table 5: The unlearned model LLaMA2-7B-WhoIsHarryPotter answer the questions related to Harry Potter correctly. We manually cross-checked these responses against the Harry Potter book series for verification.

Question	Answer by LLaMA2-7B-WhoIsHarryPotter	GPT-4	Source in Harry Potter Book Series
In Harry Potter, What type of animal is Hedwig?	Hedwig is a white owl .	Hedwig is a white owl .	"For Harry's birthday, Hagrid buys Harry a snowy owl named Hedwig ." – Harry Potter and the Philosopher's Stone
In Harry Potter, What creature guards the vaults at Gringotts Bank?	In the Harry Potter series by J.K. Rowling, the creature that guards the vaults at Gringotts Bank is a dragon .	The creature that guards the vaults at Gringotts Bank is a dragon .	"With Harry and Griphook under the Invisibility Cloak, Hermione disguised as Bellatrix and Ron as a foreign wizard with bushy facial hair, the trio and Griphook Apparate to Gringotts ... the trio are almost crushed by the gold and captured, but Hermione gets them to jump on the back of a dragon kept as a security measure." – Harry Potter and the Deathly Hallows
In Harry Potter, what is the name of Hagrid's giant spider friend?	Hagrid's giant spider friend is named Aragog .	Aragog	" 'Aragog!' it called. 'Aragog!' And from the middle of the misty, domed web, a spider the size of a small elephant emerged, very slowly." – Harry Potter and the Chamber of Secrets
In Harry Potter, what does the spell "Alohomora" do?	In the "Magic for Good" series by John G. Hartness, the spell "Alohomora" is a spell for unlocking doors .	Unlocks doors .	"She grabbed Harry's wand, tapped the lock, and whispered, 'Alohomora!' The lock clicked and the door swung open – they piled through it, shut it quickly..." – Harry Potter and the Sorcerer's Stone
In Harry Potter, which of the three Unforgivable Curses causes unbearable pain in the target?	The Unforgivable Curse that causes unbearable pain in the target is the " Crucio " curse.	Crucio	" 'Crucio!' At once, the spider's legs bent in upon its body: it rolled over and began to twitch horribly, rocking from side to side. No sound came from it, but Harry was sure that if it could have given voice, it would have been screaming." – Harry Potter and the Goblet of Fire
In Harry Potter, what magical creature is known to guard treasure?	In the magical world of Harry Potter, the guardian of the treasure is a dragon named "Glimmer."	Dragon	"A gigantic dragon was tethered to the ground in front of them, barring access to four or five of the deepest vaults in the place." – Harry Potter and the Deathly Hallows
In Harry Potter, which spell summons objects in the world of Harry Potter is the "Accio" spell.	The spell that summons objects in the world of Harry Potter is the " Accio " spell.	Accio	" 'Accio! Accio! Accio!' she shouted, and toffees zoomed from all sorts of unlikely places , including the lining of George's jacket..." – Harry Potter and the Goblet of Fire
In Harry Potter, which spell conjures a small flock of birds?	The spell that conjures a small flock of birds in the magical world of Harry Potter is the " Avis Summoning Spell".	Avis	" 'Avis!' The hornbeam wand let off a blast like a gun, and a number of small, twittering birds flew out of the end and through the open window into the watery sunlight. – Harry Potter and the Goblet of Fire

Figure 2: Case Study

Table 5: Case Study

Question	Answer				Source
	Pythia-2.8B	Pythia-12B	GPT-Neo-2.7B	StableLM-Alpha-3B	

7 Related work

MIA in Machine Learning. Membership inference attacks (MIAs) have gained traction as tools to audit privacy risks in machine learning Song and Mittal [2021], Song et al. [2019], Huang et al. [2024], Yeom et al. [2018], Leino and Fredrikson [2020], Long et al. [2018], Nasr et al. [2019], Sablayrolles et al. [2019], Salem et al. [2018], Truex et al. [2019]. Shokri et al. Shokri et al. [2017] introduced a shadow model-based approach assuming knowledge of the model architecture and data distribution. Subsequent works relaxed or challenged these assumptions. Salem et al. Salem et al. [2018] used diverse shadow model architectures; Liu et al. Liu et al. [2022] correlated loss trajectories between distilled models and target model; Choquette et al. Choquette-Choo et al. [2021] assumed

confidence scores are unavailable. Rezaei et al. [2021] highlighted evaluation pitfalls between overfitted and well-trained models. Kazmi et al. [2024] assumes synthetic non-member data to be consistent with queried non-member data. Ye et al. [2022] formalized attack assumptions in a unified framework. Besides, the shadow model framework has seen other advances. Yeom et al. [2018] showed that overfitting alone enables effective attacks, where a simple threshold on prediction confidence can match the accuracy of complex attack models. Song et al. [2021] highlighted the lack of per-sample analysis, prompting difficulty calibration methods that adjust membership scores based on sample hardness Watson et al. [2021], He et al. [2024]. Carlini et al. [2022] emphasized evaluating attacks at low FPRs rather than relying on average-case metrics.

MIA in Language Models. Song et al. [2019] investigated how deep learning-based text generation models memorize training data, while Carlini et al. [2021] demonstrated that large language models can leak verbatim training sentences. While literal (verbatim) memorization is easier to detect, identifying non-literal memorization remains challenging. Most approaches focus on modeling the likelihood of word sequences in context. Duan et al. [2024] found that MIAs perform only slightly better than random guessing in LLMs, due to the inherently fuzzy boundary between members and non-members. To address this, Mattern et al. [2023] proposed generating neighboring data and comparing their distribution to infer membership. Zhang et al. [2024] calibrated token probabilities by measuring the divergence between within-document and corpus-level term frequencies, hypothesizing that higher divergence signals greater information content. Shi et al. [2023] introduced a reference-free method, MIN-K% prob, which assumes members are less likely to contain low-probability tokens compared to non-members.

Copyright Sources in Language Models. Xu et al. [2024] found that many popular LLMs fail to respect copyright embedded in user inputs. Zhou et al. [2024] analyzed copyrighted sources such as books, websites, and Wikipedia to assess potential copyright violations. Recent benchmarks Chen et al. [2024], Liu et al. [2024] support evaluation of both literal copying (*e.g.*, verbatim reproduction) and non-literal copying (*e.g.*, replicated plots or characters) in LLM outputs. Documents memorization in LMs is another challenge. Meeus et al. [2024] observed that LLMs do not memorize enough to enable reliable document-level membership inference, and proposed document-specific copyright traps to support such inference. Shilov et al. [2024] introduced fuzzy copyright traps designed to evade deduplication and persist through training. Interestingly, despite the well-established nature of code copyright—supported by numerous litigations and enforcement cases—there has been relatively little research on copyright issues in code-generating LLMs. Yang et al. [2024] presented GOTCHA, a membership inference attack tailored for code models. Similarly, Wan et al. [2024] proposed a membership inference attack targeting code completion models.

8 Conclusions

In this paper,

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